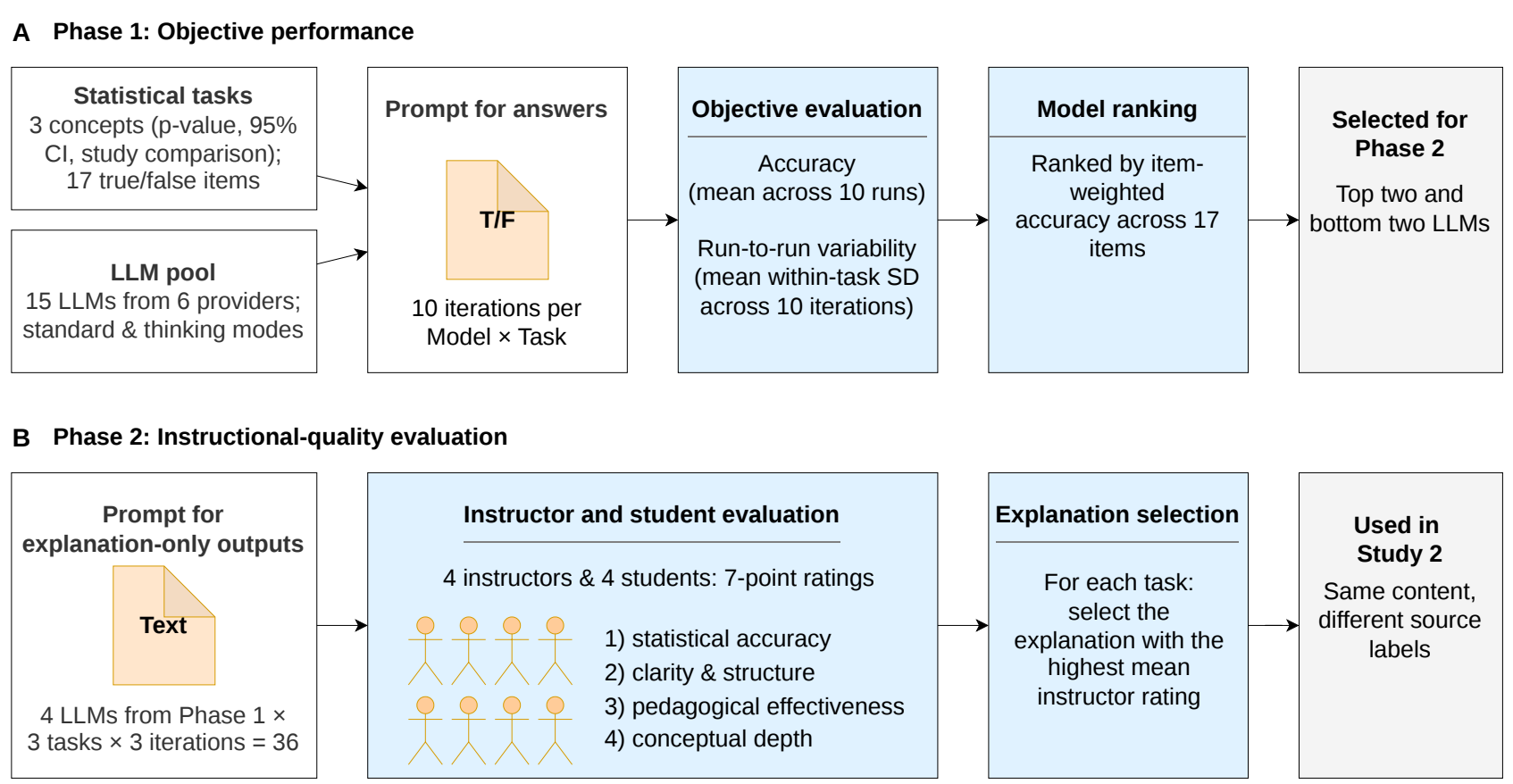
A Phase 1: Objective performance
Statistical tasks
3 concepts (p-value, 95% CI, study comparison); 17 true/false items
LLM pool
15 LLMs from 6 providers; standard & thinking modes
Prompt for answers
T/F
10 iterations per Model × Task
Objective evaluation
Accuracy (mean across 10 runs)
Run-to-run variability (mean within-task SD across 10 iterations)
Model ranking
Ranked by item-weighted accuracy across 17 items
Selected for Phase 2
Top two and bottom two LLMs
B Phase 2: Instructional-quality evaluation
Prompt for explanation-only outputs
Text
4 LLMs from Phase 1 × 3 tasks × 3 iterations = 36
Instructor and student evaluation
4 instructors & 4 students: 7-point ratings
1) statistical accuracy
2) clarity & structure
3) pedagogical effectiveness
4) conceptual depth
Explanation selection
For each task: select the explanation with the highest mean instructor rating
Used in Study 2
Same content, different source labels